

DAY 25 / 30

SUMMIT RIDGE · ML ENGINEERING

Overfitting vs underfitting: the balancing act

Overfitting vs underfitting: the balancing act

The central tension in every model you'll ever train:
performing well on data it's already seen versus performing
well on data it hasn't.

- ◆ Overfitting: the model memorizes the training data — including its noise and quirks — instead of learning the underlying pattern. It scores great on training data and disappoints on anything new. Like a student who memorizes last year's exam answers instead of understanding the subject — perfect on that exact test, lost on a new one.
- ◆ Underfitting: the model is too simple to capture the real pattern at all, and performs poorly even on the training data. Like trying to fit a straight line to a curve — it misses the shape no matter how much data you give it.

“If a model's training score and real-world score tell wildly different stories, that gap itself is a diagnosis: it's telling you the model learned the wrong thing.”

Follow along – Day 26 drops tomorrow.

#MachineLearning

#DataScience

#AI

#ModelTraining